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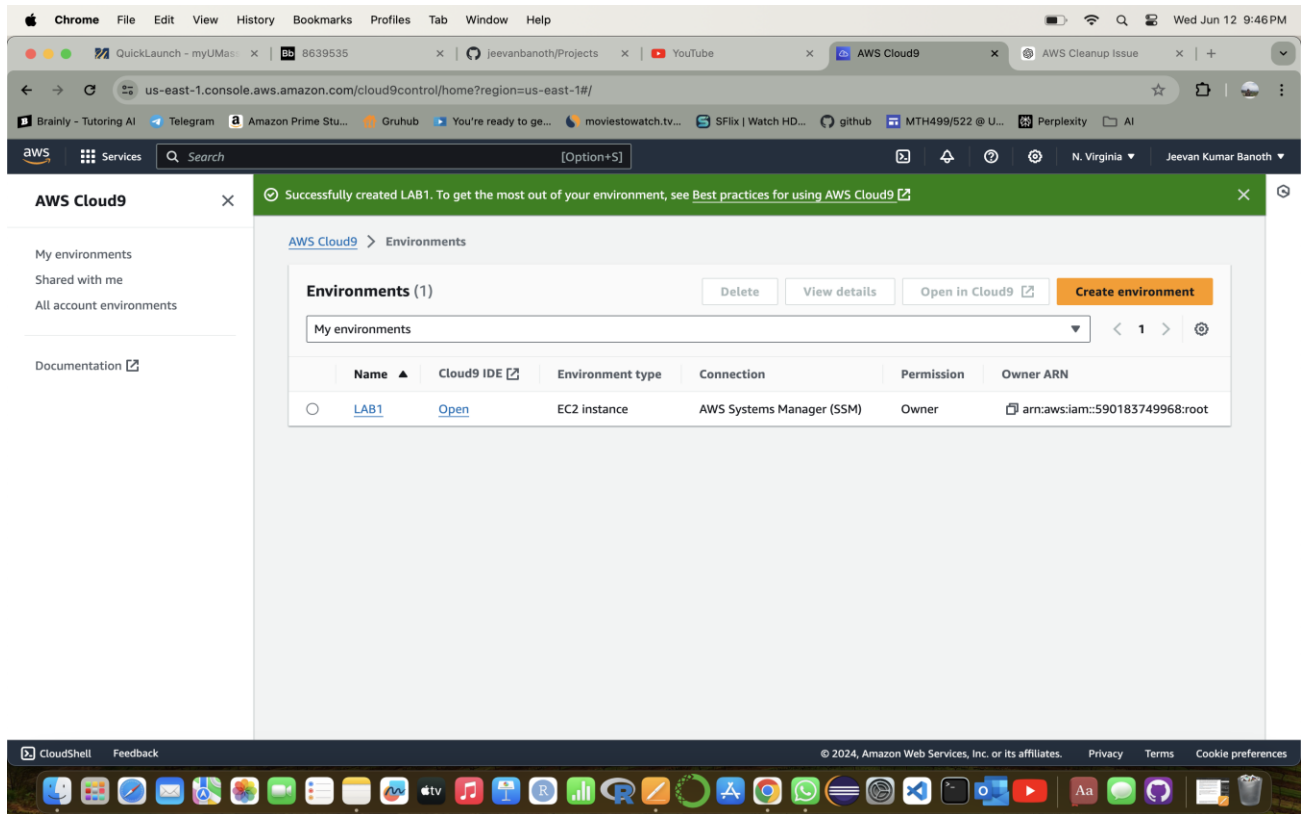
CIS 602-7103: Special Topics in CIS

Homework – 1:

Accessing and Analyzing Data by Using Amazon S3: -

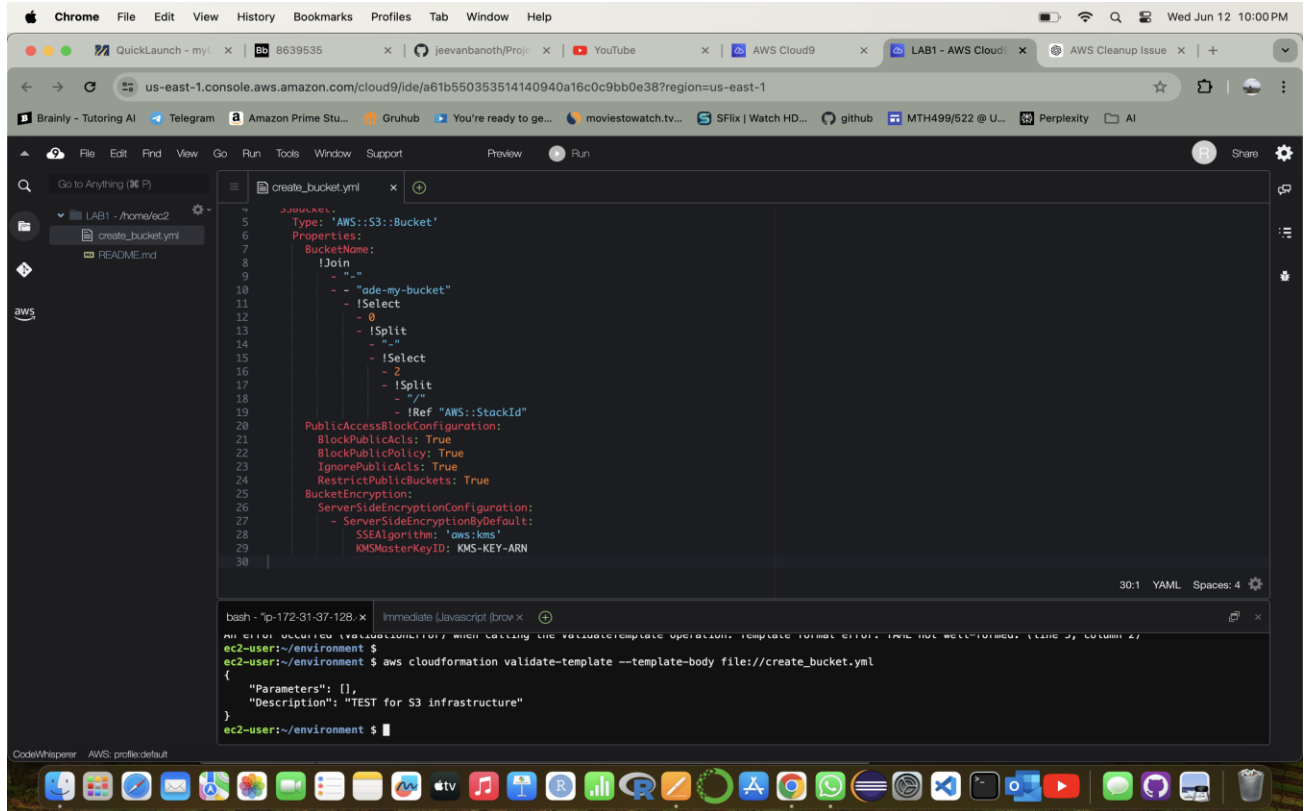
Task 1: Creating a CloudFormation template and stack

→ I tried accessing the AWS academy student version but due to some region issues the Lab is not starting in there so I will be doing my Lab in the AWS academy that I created on my own.



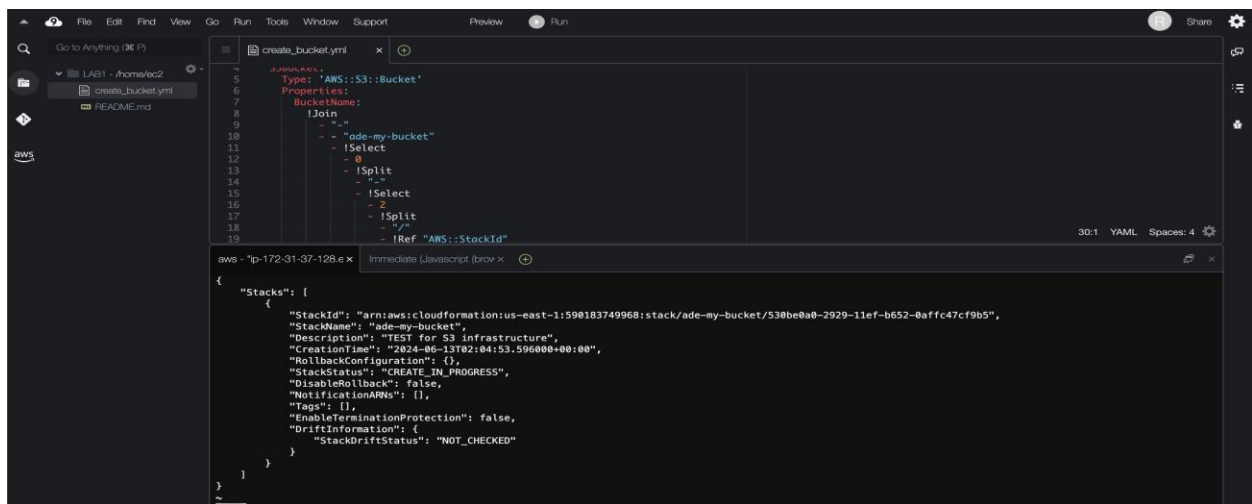
Step 1: Set Up AWS Cloud9 Environment: -

By using the Cloud9 formation in the AWS, we will be creating an .yaml template file and saving the file as create_bucket.yaml and running the commands which are given.

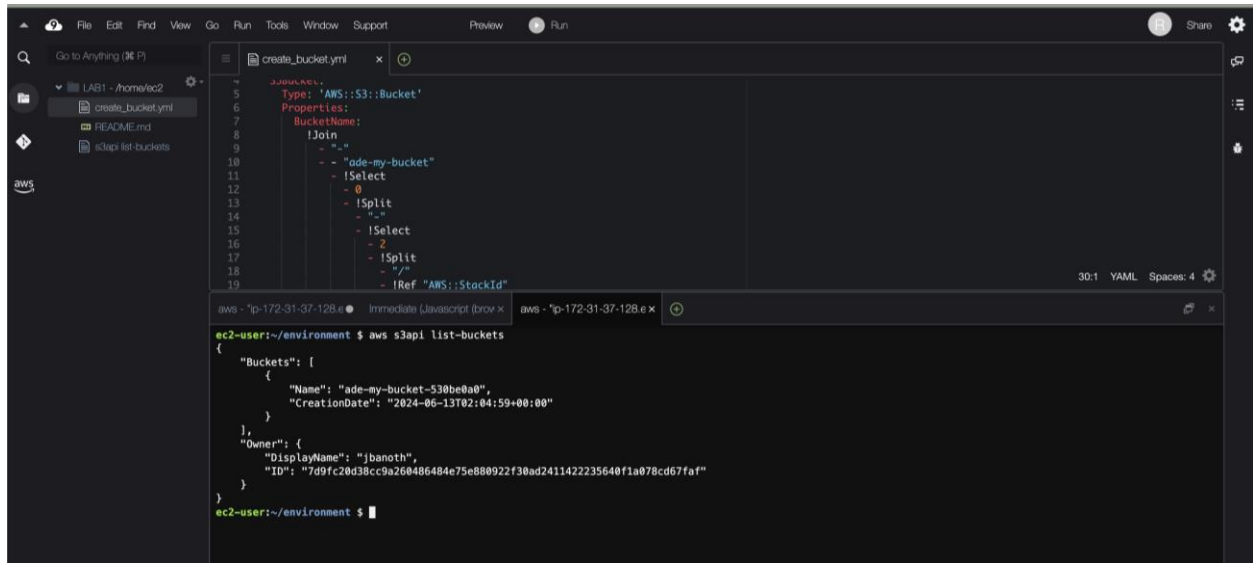


By using the commands given in the manual file, I have validated the in the AWS cloud9 formation template as shown above and it shows that template has been validated correctly.

Next, we will be creating the stack using the command given in the manual and replacing it with the stack name below.



As we can see over there, stack has been created successfully. I have also used the commands which verify that stack has been created or not which are shown in the below image.

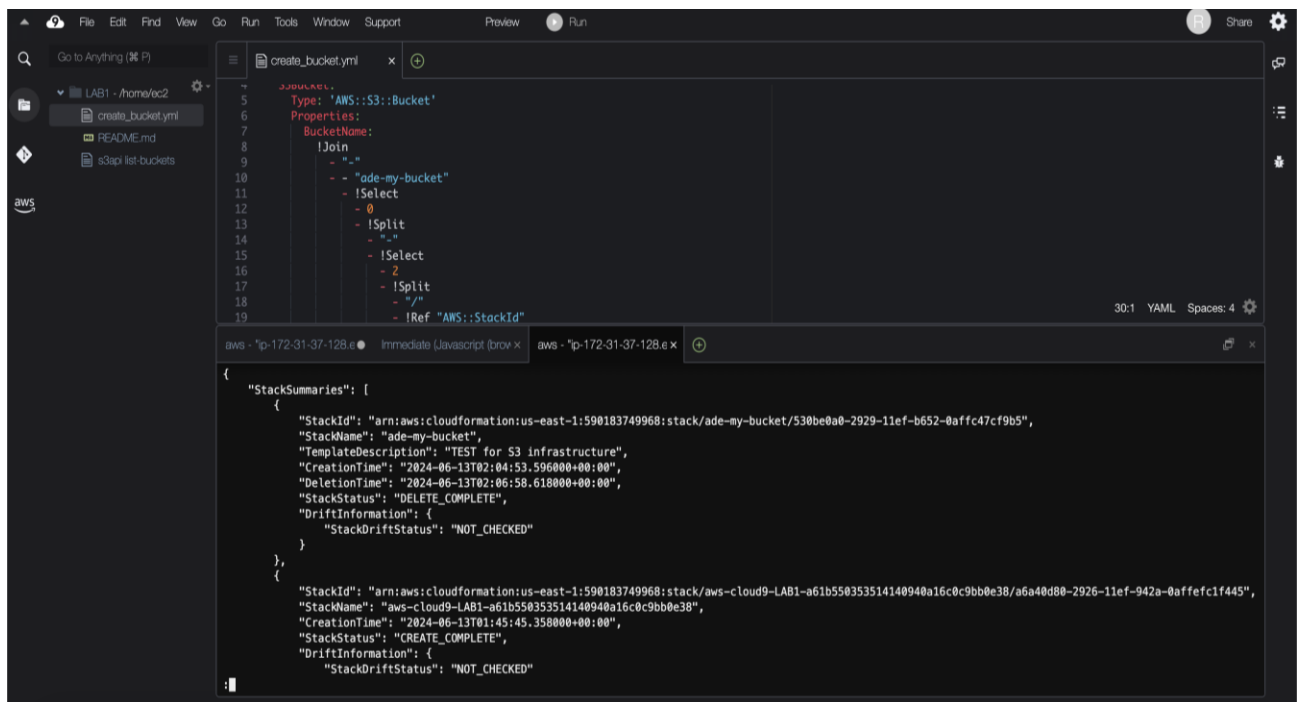


The screenshot shows a VS Code editor with a file explorer on the left containing 'create_bucket.yml', 'README.md', and 's3api list-buckets'. The main editor displays the 'create_bucket.yml' template, which defines an S3 bucket named 'ade-my-bucket'. Below the editor, a terminal window shows the command 'aws s3api list-buckets' being executed, resulting in a JSON response that lists the bucket 'ade-my-bucket' with its creation date and owner information.

```
YAML
1
2
3
4
5 Type: 'AWS::S3::Bucket'
6 Properties:
7   BucketName:
8     !Join
9       - "-"
10      - "ade-my-bucket"
11   !Select
12     - 0
13   !Split
14     - "+"
15   !Select
16     - 2
17   !Split
18     - "/"
19   !Ref "AWS::StackId"
```

```
aws - "ip-172-31-37-128.e" Immediate (Javascript (brow x) aws - "ip-172-31-37-128.e x
ec2-user:~/environment $ aws s3api list-buckets
{
  "Buckets": [
    {
      "Name": "ade-my-bucket-530be0a0",
      "CreationDate": "2024-06-13T02:04:59+00:00"
    }
  ],
  "Owner": {
    "DisplayName": "jbanoth",
    "ID": "7d9fc20d38cc9a260486484e75e880922f30ad2411422235640f1a078cd67faf"
  }
}
ec2-user:~/environment $
```

Now, by using the new command tab, I have deleted the stack using the command prompt which is given below, and verified whether that stack has been deleted successfully or not and that it came out as below one.



The screenshot shows the same VS Code editor with the 'create_bucket.yml' template. The terminal window now displays the output of the 'aws cloudformation delete-stack' command, showing a JSON response with 'StackSummaries'. The response indicates that the stack 'ade-my-bucket' has been successfully deleted, with a status of 'DELETE_COMPLETE'.

```
YAML
1
2
3
4
5 Type: 'AWS::S3::Bucket'
6 Properties:
7   BucketName:
8     !Join
9       - "-"
10      - "ade-my-bucket"
11   !Select
12     - 0
13   !Split
14     - "+"
15   !Select
16     - 2
17   !Split
18     - "/"
19   !Ref "AWS::StackId"
```

```
aws - "ip-172-31-37-128.e" Immediate (Javascript (brow x) aws - "ip-172-31-37-128.e x
{
  "StackSummaries": [
    {
      "StackId": "arn:aws:cloudformation:us-east-1:590183749968:stack/ade-my-bucket/530be0a0-2929-11ef-b652-0affc47cf9b5",
      "StackName": "ade-my-bucket",
      "TemplateDescription": "TEST for S3 infrastructure",
      "CreationTime": "2024-06-13T02:04:53.596000+00:00",
      "DeletionTime": "2024-06-13T02:06:58.618000+00:00",
      "StackStatus": "DELETE_COMPLETE",
      "DriftInformation": {
        "StackDriftStatus": "NOT_CHECKED"
      }
    },
    {
      "StackId": "arn:aws:cloudformation:us-east-1:590183749968:stack/aws-cloud9-LAB1-a61b550353514140940a16c0c9bb0e38/a6a40d80-2926-11ef-942a-0affec1f445",
      "StackName": "aws-cloud9-LAB1-a61b550353514140940a16c0c9bb0e38",
      "TemplateDescription": "2024-06-13T01:45:45.358000+00:00",
      "CreationTime": "2024-06-13T01:45:45.358000+00:00",
      "StackStatus": "CREATE_COMPLETE",
      "DriftInformation": {
        "StackDriftStatus": "NOT_CHECKED"
      }
    }
  ]
}
```

As the cursor has been stuck over there, I will be using the new terminal to check how many buckets are there.

```
aws - "ip-172-31-37-128.e" Immediate (javascript (brow x aws - "ip-172-31-37-128.e x AWS - "ip-172-31-37-128.e x
ec2-user:~/environment $ aws s3api list-buckets

{
  "Buckets": [],
  "Owner": {
    "DisplayName": "jbanoth",
    "ID": "7d9fc20d38cc9a260486484e75e880922f30ad2411422235640f1a078cd67faf"
  }
}

ec2-user:~/environment $
ec2-user:~/environment $
```

As we have deleted all my stack which I have created just now, the list in the bucket shows empty.

SUMMARY OF THE TASK 1:

So, in this task we have created cloud9 environment template as given and developed a template for an S3 bucket.

We have validated the template using AWS CLI.

Deployed the stack using terminal window in cloud9 and verified the S3 bucket.

Also, by using the same terminal we have deleted the stack which is created and verified if the stack has been successfully deleted or not.

Next, we checked how many buckets are present in the environment and summarized it.

Task 2: Uploading sample data to an S3 bucket: -

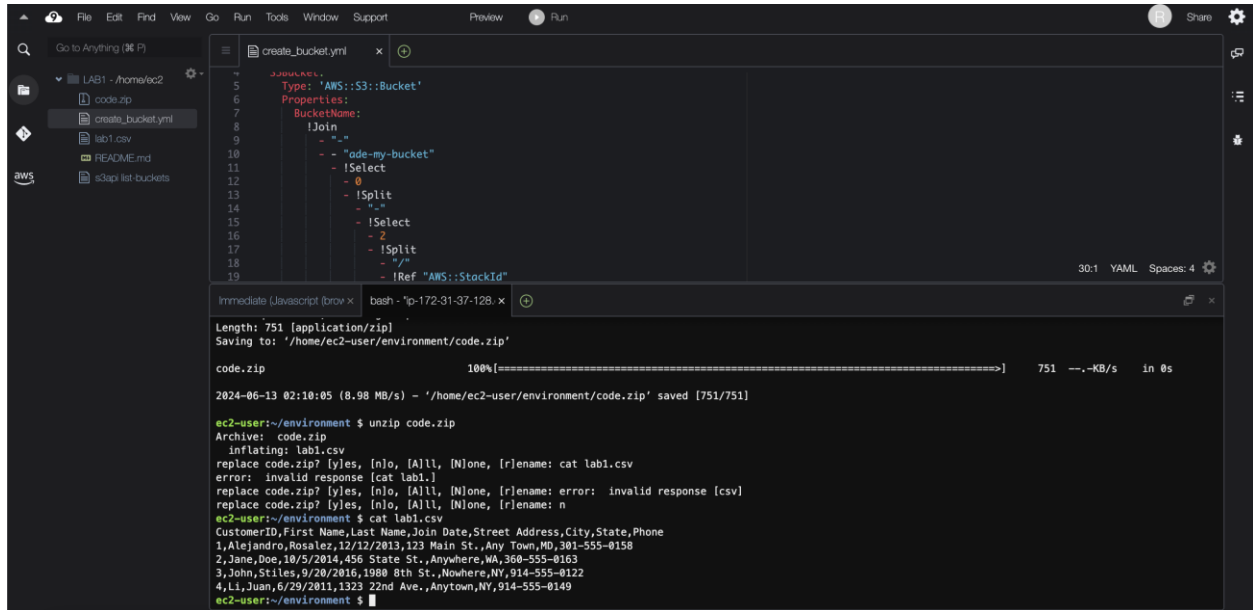
As we have deleted all the stacks which have been created, we will be recreating it by using the same command which is used before.

```
Immediate (javascript (brow x aws - "ip-172-31-37-128.e x
ec2-user:~/environment $ aws cloudformation describe-stacks --stack-name ade-my-bucket
{
  "Stacks": [
    {
      "StackId": "arn:aws:cloudformation:us-east-1:590183749968:stack/ade-my-bucket/f6147af0-2929-11ef-a294-12af708fdd03",
      "StackName": "ade-my-bucket",
      "Description": "TEST for S3 infrastructure",
      "CreationTime": "2024-06-13T02:09:27.129000+00:00",
      "RollbackConfiguration": {},
      "StackStatus": "CREATE_IN_PROGRESS",
      "DisableRollback": false,
      "NotificationARNs": [],
      "Tags": [],
      "EnableTerminationProtection": false,
      "DriftInformation": {
        "StackDriftStatus": "NOT_CHECKED"
      }
    }
  ]
}
```

In cloud9 terminal, we have used the given commands for downloading and unzipping the given file as below.

When prompted to unzip the code.zip file we need to enter n and return it.

To view the contents which are present in the file, we will be using the cat command and as shown below.



```
create_bucket.yml
1
2
3
4
5 Type: 'AWS::S3::Bucket'
6
7 Properties:
8   BucketName:
9     !Join
10      - "-"
11      - !Select
12        - 0
13      - !Split
14        - "-"
15      - !Select
16        - 2
17      - !Split
18        - "/"
19      - !Ref "AWS::StackId"
```

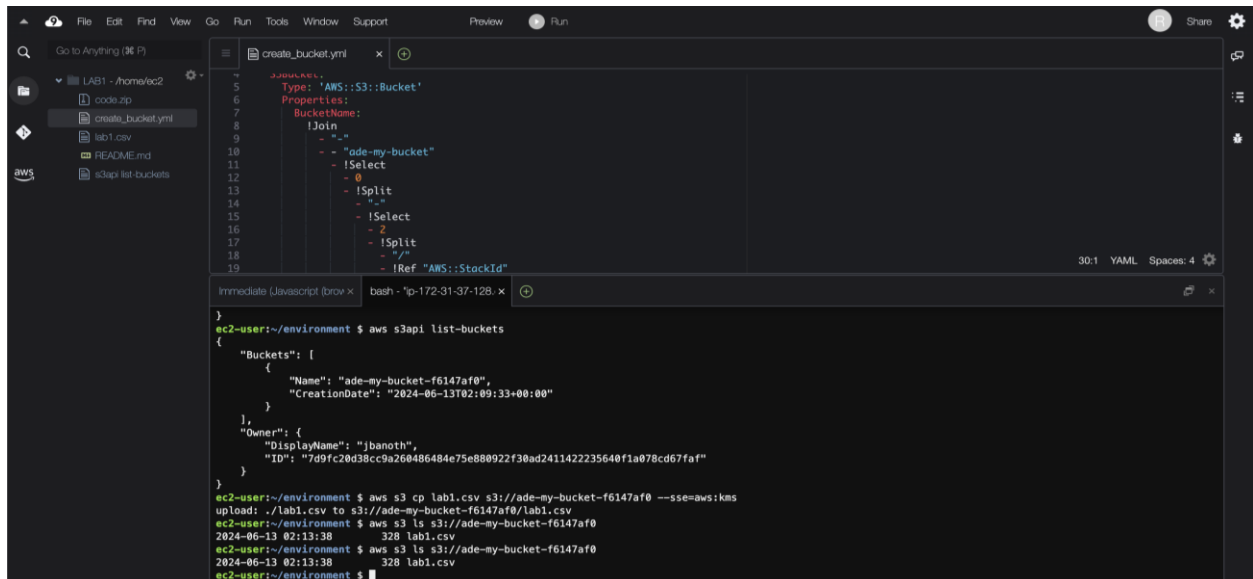
```
Immediate (JavaScript (browser) x bash - 'ip-172-31-37-128.x
Length: 751 [application/zip]
Saving to: '/home/ec2-user/environment/code.zip'

code.zip
100%[=====] 751 --KB/s in 0s

2024-06-13 02:10:05 (8.98 MB/s) - '/home/ec2-user/environment/code.zip' saved [751/751]

ec2-user:~/environment $ unzip code.zip
Archive: code.zip
  inflating: lab1.csv
replace code.zip? [yes, [n]o, [A]ll, [N]one, [r]ename: cat lab1.csv
error: invalid response [cat lab1.]
replace code.zip? [yes, [n]o, [A]ll, [N]one, [r]ename: error: invalid response [csv]
replace code.zip? [yes, [n]o, [A]ll, [N]one, [r]ename: n
ec2-user:~/environment $ cat lab1.csv
CustomerID,First Name,Last Name,Join Date,Street Address,City,State,Phone
1,Alejandro,Rosalez,12/12/2013,123 Main St.,Any Town,MD,301-555-0158
2,Jane,Doe,10/5/2014,456 State St.,Anywhere,WA,360-555-0163
3,John,Stiles,9/20/2015,1900 8th St.,Nowhere,NY,914-555-0122
4,Li,Juan,6/29/2011,1323 22nd Ave.,Anytown,NV,514-555-0149
ec2-user:~/environment $
```

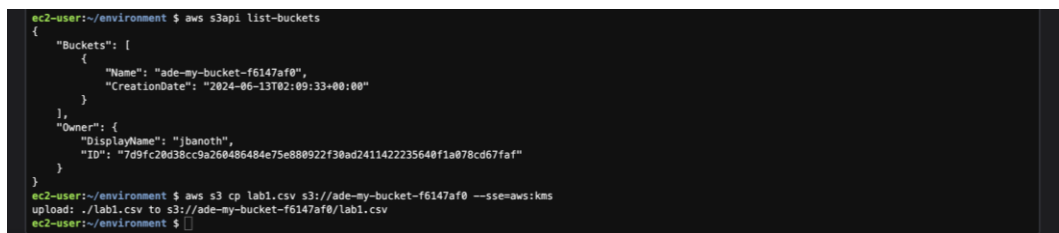
Next, we will be listing all the buckets which are present. And copying the dataset to S3 bucket. Then verifying the upload is successful or not.



```
create_bucket.yml
1
2
3
4
5 Type: 'AWS::S3::Bucket'
6
7 Properties:
8   BucketName:
9     !Join
10      - "-"
11      - !Select
12        - 0
13      - !Split
14        - "-"
15      - !Select
16        - 2
17      - !Split
18        - "/"
19      - !Ref "AWS::StackId"
```

```
ec2-user:~/environment $ aws s3api list-buckets
{
  "Buckets": [
    {
      "Name": "ade-my-bucket-f6147af0",
      "CreationDate": "2024-06-13T02:09:33+00:00"
    }
  ],
  "Owner": {
    "DisplayName": "jbanoth",
    "ID": "7d9fc20d38cc9a260486484e75e880922f30ad2411422235640f1a078cd67faf"
  }
}

ec2-user:~/environment $ aws s3 cp lab1.csv s3://ade-my-bucket-f6147af0 --sse=aws:kms
upload: ./lab1.csv to s3://ade-my-bucket-f6147af0/lab1.csv
ec2-user:~/environment $ aws s3 ls s3://ade-my-bucket-f6147af0
2024-06-13 02:13:38          328 lab1.csv
ec2-user:~/environment $ aws s3 ls s3://ade-my-bucket-f6147af0
2024-06-13 02:13:38          328 lab1.csv
ec2-user:~/environment $
```



```
ec2-user:~/environment $ aws s3api list-buckets
{
  "Buckets": [
    {
      "Name": "ade-my-bucket-f6147af0",
      "CreationDate": "2024-06-13T02:09:33+00:00"
    }
  ],
  "Owner": {
    "DisplayName": "jbanoth",
    "ID": "7d9fc20d38cc9a260486484e75e880922f30ad2411422235640f1a078cd67faf"
  }
}

ec2-user:~/environment $ aws s3 cp lab1.csv s3://ade-my-bucket-f6147af0 --sse=aws:kms
upload: ./lab1.csv to s3://ade-my-bucket-f6147af0/lab1.csv
ec2-user:~/environment $
```

Task 2 summary: -

In this task, we have recreated S3 bucket using CloudFormation template.

Downloaded and unzipped the sample data file which is there. And examined the data in the csv file which is unzipped.

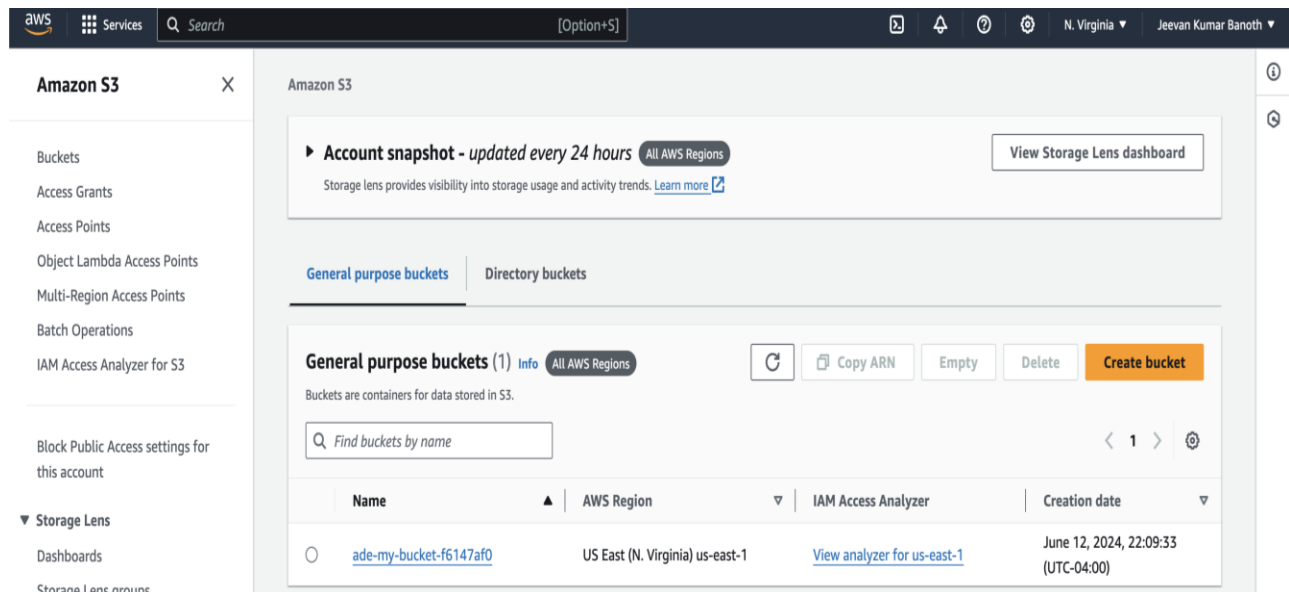
Then I used AWS CLI commands to copy the dataset file which is downloaded into the S3 bucket.

Next, we verified if the copying dataset into the S3 bucket was done successfully or not.

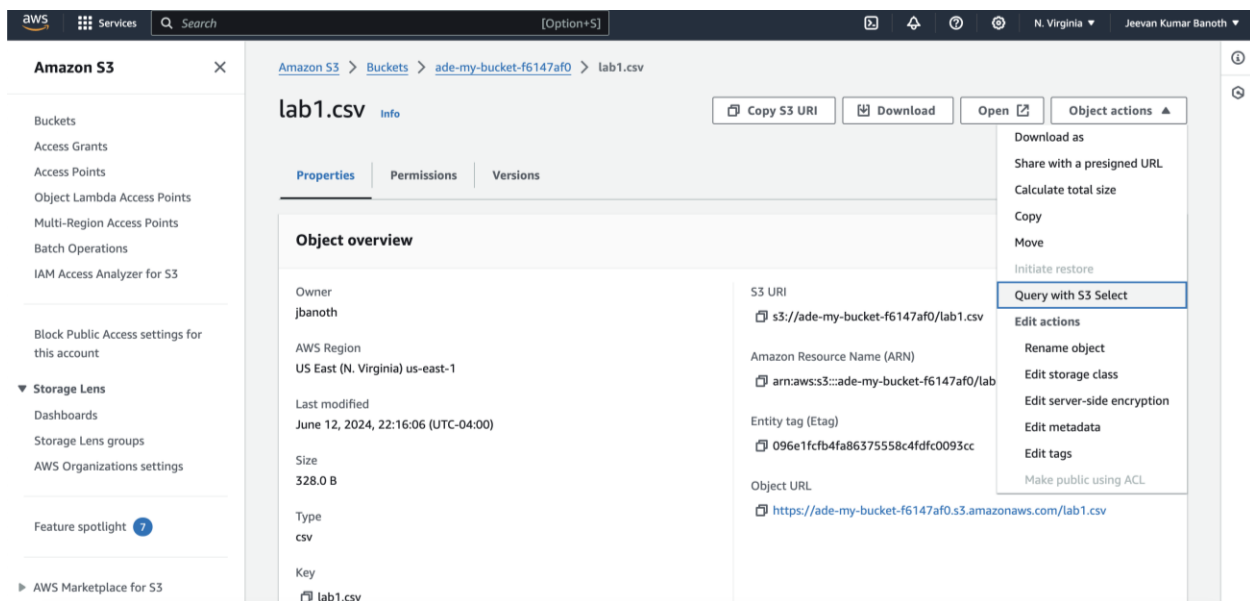
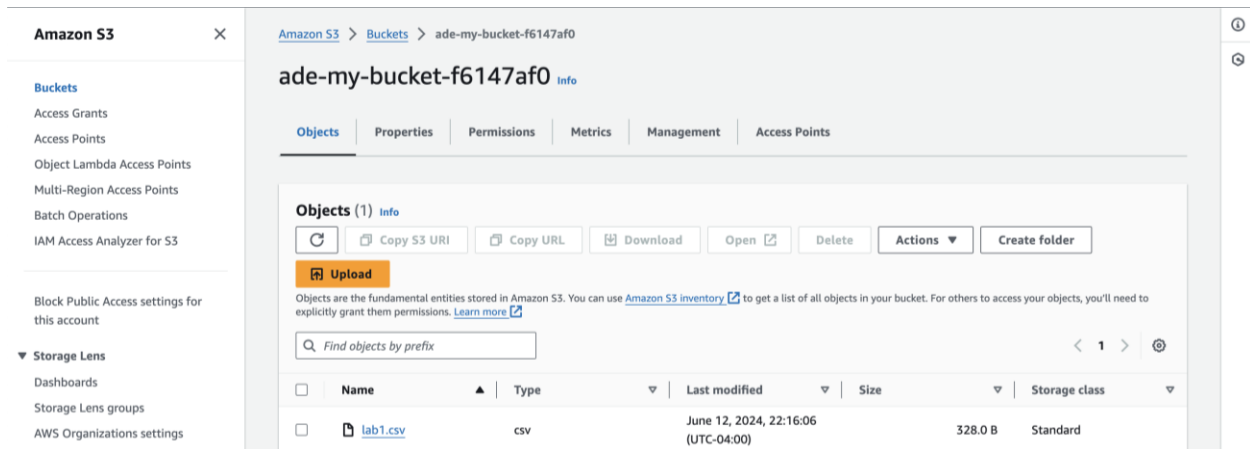
This exercise will clarify how to use Amazon S3 bucket and perform small operations like creating the stack and managing the dataset.

Task 3: Querying the data with S3 select: -

Now, we will be accessing the S3 bucket in AWS management console for that first, we open the AWS management console and open the S3 bucket from the search bar, in the bucket list, we will be opening the existing bucket.



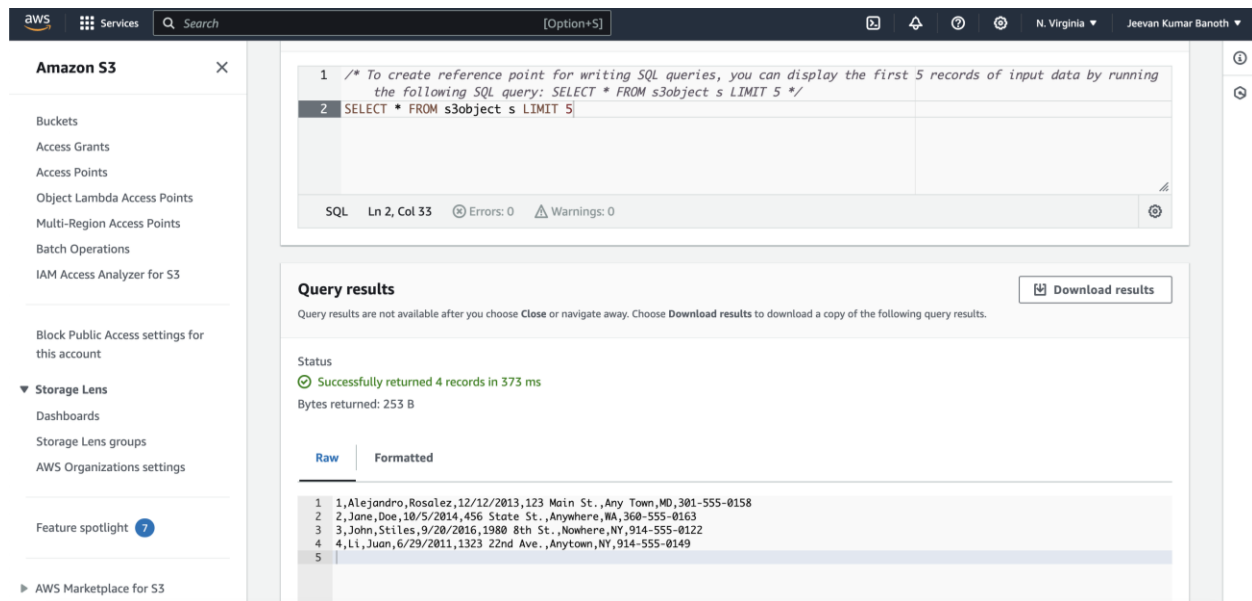
Inside the bucket, I will be selecting the data file which is lab1.csv, next in the Object actions choosing Query with S3 select.



After this, in the input settings section we will be keeping the format as CSV selected. In the CSV delimiter keep the comma as selected and select the exclude the first line of the CSV data.

For compression keeping none selected.

For output configure setting keeping the default values as CSV and comma selected.



The screenshot shows the AWS S3 console interface. On the left is a navigation pane with options like Buckets, Access Grants, Access Points, Object Lambda Access Points, Multi-Region Access Points, Batch Operations, IAM Access Analyzer for S3, Block Public Access settings for this account, Storage Lens (Dashboards, Storage Lens groups, AWS Organizations settings), and Feature spotlight. The main area is titled 'Amazon S3' and contains a SQL editor and a 'Query results' section.

SQL Editor:

```
1 /* To create reference point for writing SQL queries, you can display the first 5 records of input data by running
2 the following SQL query: SELECT * FROM s3object s LIMIT 5 */
3 SELECT * FROM s3object s LIMIT 5
```

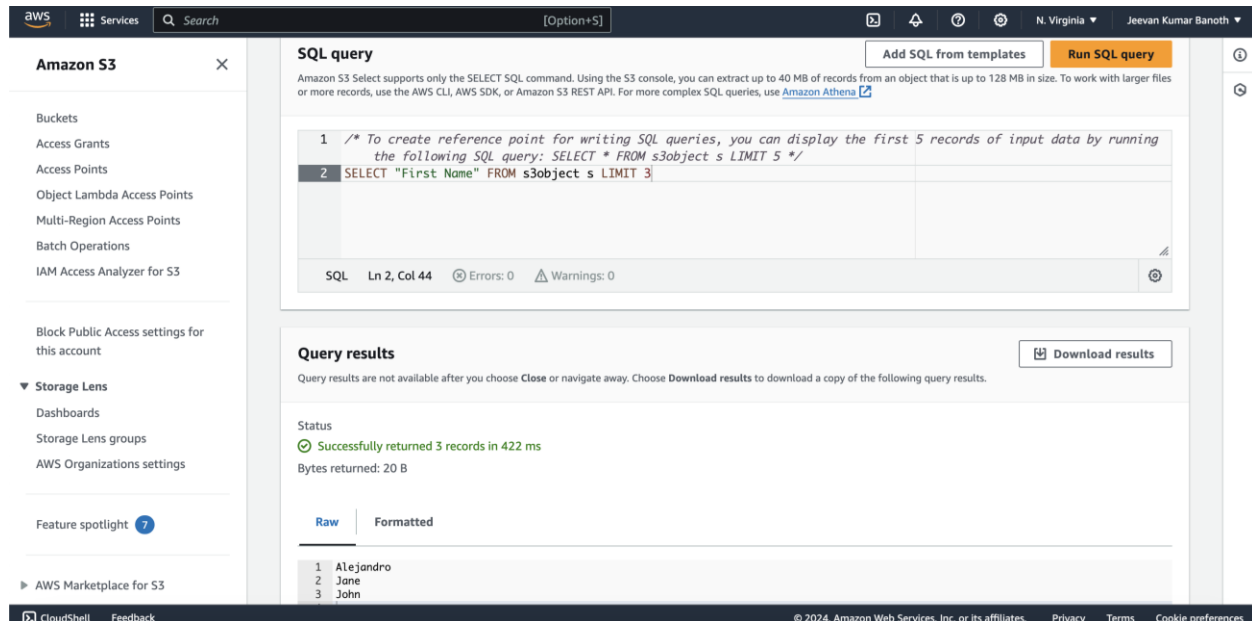
Query results:

Status: Successfully returned 4 records in 373 ms
Bytes returned: 253 B

Raw | Formatted

1	,Alejandro,Rosalez,12/12/2013,123 Main St.,Any Town,MD,301-555-0158
2	2,Jane,Doe,10/5/2014,456 State St.,Anywhere,IA,360-555-0163
3	3,John,Stiles,9/20/2016,1980 8th St.,Nowhere,NY,914-555-0122
4	4,Li,Juan,6/29/2011,1323 22nd Ave.,Anytown,NY,914-555-0149
5	

As shown above we need to run the SQL Query “SELECT * FROM s3object s LIMIT 5” as this, which will give all rows present in there.



The screenshot shows the AWS S3 console interface with a modified SQL query. The query is 'SELECT "First Name" FROM s3object s LIMIT 3'. The results show 3 records in a CSV format, displaying only the first names.

SQL Editor:

```
1 /* To create reference point for writing SQL queries, you can display the first 5 records of input data by running
2 the following SQL query: SELECT * FROM s3object s LIMIT 5 */
3 SELECT "First Name" FROM s3object s LIMIT 3
```

Query results:

Status: Successfully returned 3 records in 422 ms
Bytes returned: 20 B

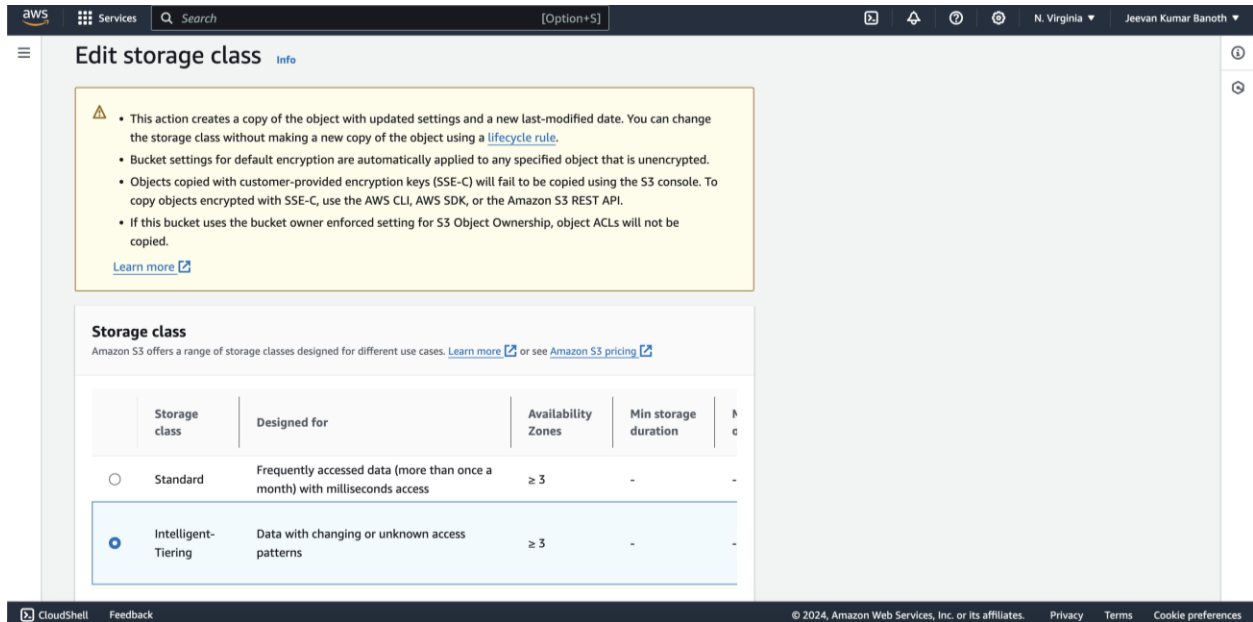
Raw | Formatted

1	Alejandro
2	Jane
3	John

First, we will be excluding the first line of CSV data that is selected in the Input Settings section. Then we will be changing the SQL Query in such a way that it will display only the first name of the people and we have limited it to only 3 rows.

Task 4: Modifying an Object's Encryption Properties and Storage Type: -

Now we will be modifying the object encryption properties by storage type in the edit section by choosing Intelligent-Tiering.



Edit storage class [Info](#)

Warning:

- This action creates a copy of the object with updated settings and a new last-modified date. You can change the storage class without making a new copy of the object using a [lifecycle rule](#).
- Bucket settings for default encryption are automatically applied to any specified object that is unencrypted.
- Objects copied with customer-provided encryption keys (SSE-C) will fail to be copied using the S3 console. To copy objects encrypted with SSE-C, use the AWS CLI, AWS SDK, or the Amazon S3 REST API.
- If this bucket uses the bucket owner enforced setting for S3 Object Ownership, object ACLs will not be copied.

[Learn more](#)

Storage class
Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

	Storage class	Designed for	Availability Zones	Min storage duration	Price
☐	Standard	Frequently accessed data (more than once a month) with milliseconds access	≥ 3	-	-
☒	Intelligent-Tiering	Data with changing or unknown access patterns	≥ 3	-	-

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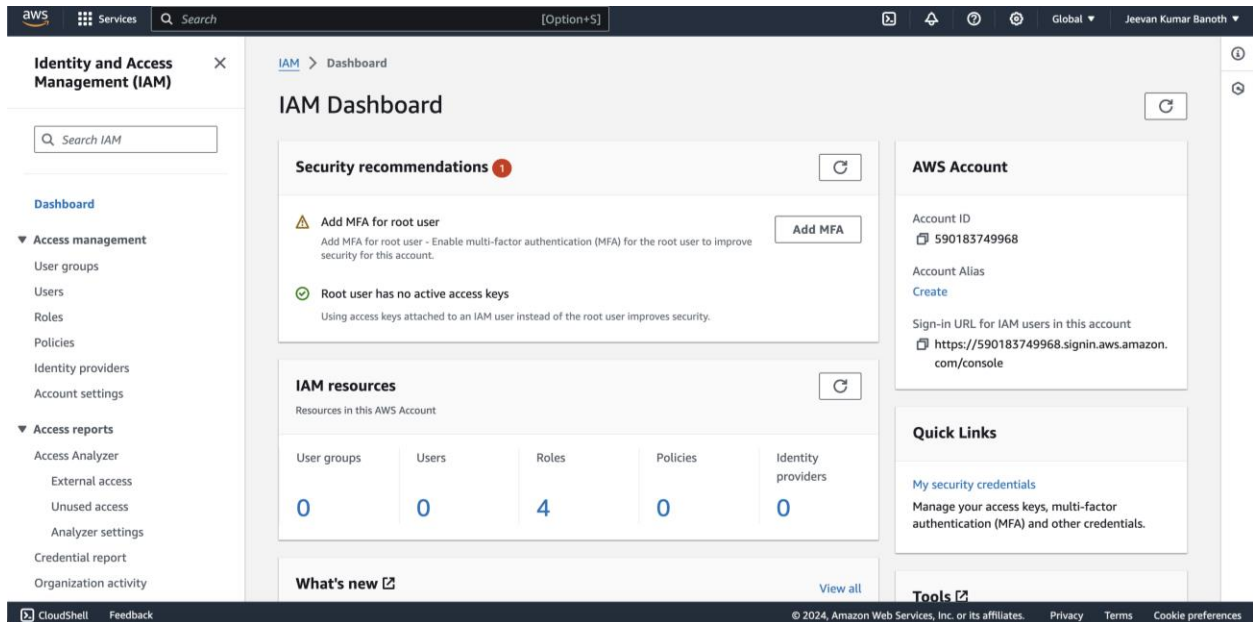
Task 5: Compressing and Querying the Dataset: -

Now, I will be compressing the data file by using AWS Cloud9 terminal and we will be zipping it. Then we will be gunzipping the zipped data file.

To list the objects in the directory, we will be using “ls -la” and running it.

Task 6:

Managing and Testing Restricted Access for a Team Member:



For performing this above task which is managing and testing restricted access for Team members, I won't be able to do it because as I am performing this from my separate account, so I don't have the access to any of the groups.

This is the reason why I won't be able to perform the task from now on.

Summary: -

So, in this lab I have learned that:

how to create an S3 bucket using a cloud9 formation template to ensure that repeatable and automated process.

Added the data into S3 Bucket and performed few operations like creating stack, deleting and listing all stacks present in there.

Copying the data from the stack into bucket and viewing the contents of the dataset by using the cat commands.

Using the SQL Query to in the S3 select to view the contents in the dataset.

I got to know how to zip and unzip the dataset, which was downloaded, and how to perform the Gzip operations in that.

Ran the SQL Queries to run the specific queries.

I know how to modify the objects' encryption properties and storage type. Compressing and Quering the dataset.

So, from this lab I have learned all the above things.

-----X-----